Li Ju

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Education

Ph.D., Scientific Computing — Sep. 2021 – Jun. 2026 (Expected)
Uppsala University, Uppsala, Sweden

M.Sc., Computational Science — Sep. 2019 – Aug. 2021
Uppsala University, Uppsala, Sweden

M.Sc., Chemometrics, *with Honors* — Sep. 2016 – Jun. 2019
University of Science and Technology of China, Hefei, China

B.Sc., Chemistry — Sep. 2012 – Jun. 2016
University of Science and Technology of China, Hefei, China

Professional Experience

Doctoral Researcher — Sep. 2021 –
Scientific Machine Learning Laboratory, Uppsala University
Advisors: Andreas Hellander, Prashant Singh

Machine Learning Engineer Intern — Jan. 2026 – Apr. 2026
Modulai, Stockholm, Sweden

Data Scientist (Part-time) — Jan. 2021 – June. 2021
Scaleout Systems AB, Uppsala, Sweden

Research Interests

- Probabilistic and generative modeling
- Uncertainty quantification for multimodal foundation models
- Post-hoc adaptation and representation geometry of pre-trained encoders
- Distributed and federated learning algorithms and systems

Selected Publications

Google Scholar: scholar.google.com/citations?user=tlXTHUEAAAAJ

Conference

Ju, L., Nautiyal, M., Vats, E., & Singh, P. (2026). Epistemic Uncertainty Quantification for Pre-trained VLMs via Riemannian Flow Matching. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026).*

Ju, L., Andersson, M., Fredriksson, S., Glöckner, E., Hellander, A., Vats, E., & Singh, P. (2025). Exploiting the Asymmetric Uncertainty Structure of Pre-trained VLMs on the Unit Hypersphere. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025).*

Ju, L., Singh, P., & Toor, S. (2021). Proactive Autoscaling for Edge Computing Systems with Kubernetes. In *Proceedings of the 14th IEEE/ACM International Conference on Utility and Cloud Computing Companion (UCC '21).*

Journal

Ju, L., Zhang, T., Toor, S., & Hellander, A. (2024). Accelerating Fair Federated Learning: Adaptive Federated Adam. *IEEE Transactions on Machine Learning in Communications and Networking*, **2**, 1017–1032.

Ju, L., Hellander, A., & Spjuth, O. (2024). Federated Learning for Predicting Compound Mechanism of Action Based on Image Data From Cell Painting. *Artificial Intelligence in the Life Sciences*, **5**.

Under Review

Nautiyal, M., **Ju, L.**, Hellander, A., Vats, E., & Singh, P. (2026). GeoFlowVLM: Geometry-Aware Joint Uncertainty for Frozen Vision-Language Embeddings. *arXiv:2605.13352*.

Nautiyal, M., **Ju, L.**, Ernfors, M., Hagland, K., Holma, V., Söderholm, M. W., Hellander, A., & Singh, P. (2026). OneFlowSBI: One Model, Many Queries for Simulation-Based Inference. *arXiv:2601.22951*.

Honors and Awards

1st Place in Europe, 2nd Place Globally — 2025
Huawei Wireless Communication Global Hackathon

2nd Place, Huawei Sweden Hackathon — 2024
Huawei Sweden

M.Sc. in Chemometrics, *with Honors* — 2016 – 2019
University of Science and Technology of China

Second-Class Freshman Scholarship — 2012
University of Science and Technology of China

Invited Talks & Presentations

Invited Talks

- Sep. 2025 — "Orthogonality in Neural Networks," Scaleout Edges, Sweden.
- Jun. 2025 — "Learning from Distributed and Heterogeneous Data," Half-Time Seminar, Department of Information Technology, Uppsala University.
- May 2024 — "Federated Learning for Predicting Compound Mechanism of Action Based on Image Data from Cell Painting," AI4Research, Uppsala University.
- Apr. 2024 — "Denoising Diffusion Probabilistic Models," Ph.D. Course on Bayesian Inference, Uppsala University.
- Dec. 2023 — "Federated Learning from the Perspective of Optimization," Centre for Interdisciplinary Mathematics, Uppsala University.

Posters

- Oct. 2024 — "Is Logit Adjustment a Free Lunch for Heterogeneous Federated Learning?" eSSENCE 2024, Sweden.
- 2022 — "Accelerating Fair Federated Learning," Swedish e-Science Academy, Sweden.

Teaching

Head Teaching Assistant — 2026
Data Engineering II (1TD076), Uppsala University

Teaching Assistant — 2023 – 2025
Data Engineering II (1TD076), Uppsala University

Guest Lecturer — 2025
Distributed and Parallel Computing (1TD070), Uppsala University

Teaching Assistant — 2021 – 2022
Data Engineering I (1TD169), Uppsala University

Supervisor — 2023 – 2026

Supervised four M.Sc. student groups on uncertainty quantification, generative modeling, and simulation-based inference, Uppsala University.

Service

Leadership

President 2024 – 2025

Uppsala Chapter of the Society for Industrial and Applied Mathematics (SIAM), Uppsala

Academic Reviewing

ICML 2026, NeurIPS 2026, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Mobile Computing, IEEE Communications Magazine.

Open-Source Software

Blades (Contributor): https://github.com/bladesteam/blades 2022 – 2023

Simulator and benchmark suite for Byzantine-robust federated learning.

FEDn (Contributor): https://github.com/scaleoutsystems/scaleout-client 2020 – 2021

Open-source federated learning framework; contributed PyTorch integration.

Technical Skills

Frameworks	PyTorch, JAX, transformers, verl, vllm, FastAPI, pthread, OpenMP, MPI, CUDA
Cloud / DevOps	Docker, Apptainer, Kubernetes, OpenStack
Programming Languages	Python, C/C++, Haskell, Erlang, Lisp/Racket
Spoken Languages	Chinese (native), English (fluent)

References

Assoc. Prof. Andreas Hellander Ph.D. advisor

Department of Information Technology, Uppsala University

andreas.hellander@it.uu.se

Assoc. Prof. Prashant Singh Co-advisor

Science for Life Laboratory & Department of Information Technology, Uppsala University

prashant.singh@it.uu.se

Assis. Prof. Ekta Vats Collaborator

Department of Information Technology, Uppsala University

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